Module 7 – Main Lecture, Part 1: “From Words to Worlds”

Hello everyone and welcome to the Module 7 lecture.

And this one might seem a little bit silly. Up until now, we've been doing some complicated things with artificial intelligence and coming out of the ethics module where everything was pretty heavy. This module is going to be much lighter by comparison. The purpose of this module is to show you how to use artificial intelligence to go beyond text and data and create images and video.

Katie, I was actually surprised by how much I have been using the image and video generator. Mostly for this course, but when you think about creating announcements or posters for events, if you're looking to increase the amount of engagement on social media, even if you're just creating a PowerPoint and want it to look a little bit nicer,

I was talking to someone this week from a small independent rural hospital in eastern Kentucky. He told me that he started using AI tools for their social media posts, and he has been able to show a direct relationship between engagement on social media and people clicking to schedule appointments with their primary care doctor through their healthcare system. So even if you're not planning a career in design or communications, you might be able to offload some of the images and branding to AI.

So here’s our game plan for this module. First, we’ll look at the "what" and the "why"—exploring some real-world examples of how professionals are using this technology for creative problem-solving. Then, we’ll get into the "how"—you'll learn the art of prompt engineering specifically for images and video, which is a very different skill than prompting for text. And finally, we'll take on discussion about the ethical issues of AI-generated media, from copyright to deepfakes.

So, what exactly *is* multimodal AI? The term might sound technical, but the concept is actually pretty straightforward. "Multimodal" just means "multiple modes." So, instead of an AI that only understands the mode of language, these models are designed to process other modes, like the pixels, colors, and shapes in an image, or the frequencies in a sound file.

It's like the difference between reading Game of Thrones and watching Game of Thrones. To understand the book, the AI just needs to know language. But to understand the show, it needs to understand light, color, motion, composition, the subtle intricacies of unspoken communication. What did that knowing look from Tyrian to his father mean?

It’s a whole different kind of intelligence.

Professionals in almost every field are using these tools to solve complex problems and accelerate creativity in truly sophisticated ways. AI is becoming an essential visual brainstorming partner.

For example, let's look at architecture. This is the world-renowned Zaha Hadid Architects who are using AI, robotics and Virtual Reality to generate hundreds of unique building designs in minutes, exploring structural and aesthetic possibilities that would take human teams weeks or months to even sketch out. The images are concepts for buildings that were created from a text prompt.

Or think about product design. Companies like Nike and Adidas are experimenting with AI to rapidly prototype new sneaker designs. They can use text prompts to instantly change materials, colors, and shapes, visualizing a new product before the first physical prototype is ever made.

It's also transforming creative fields like filmmaking and advertising. Directors and marketing teams can now create entire storyboards or mockups for an ad campaign in an afternoon. They can establish the mood, the lighting, and the character design, all with AI, allowing them to align on a visual concept before a single dollar is spent on production.

We've done the same here. I made the introductory logo for this course in about 45 minutes. And then created this clip in about 5. I think it elevates this whole production and makes it feel more professional. Companies spend a lot of time thinking about their branding and logos, and now we can too.

In the sciences, researchers can prompt an AI to create visualizations of incredibly complex concepts like protein folding or the structure of a distant nebula.

This is what it looks like when two spiral galaxies collide taken with the Hubble Space Telescope

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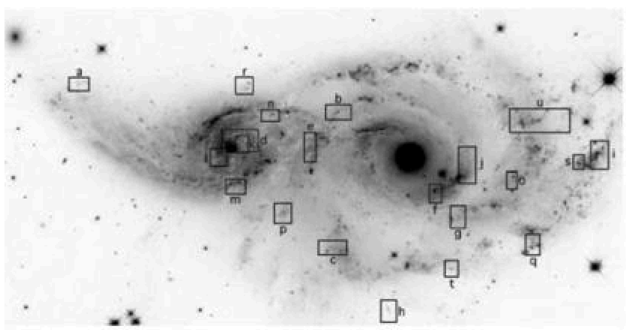

Figure 1. Grey-scale *HST B*-band image of NGC 2207 (right) and IC 2163 (left), from Paper 4. Rectangles and labels show peculiar emission features described in that paper. Previously published material reproduced by permission of the American Astronomical Society.

Here's the same image visualized with AI.

This makes abstract data visible and understandable in a way it never was before. This is AI as a tool for understanding.

So, the power and the potential are obviously huge. Now, let's get practical and talk about how you can actually do it.

Prompting for a picture is a totally different game than prompting for an essay. You are no longer having a conversation with the AI, instead you're a director on a movie set. The prompt is your script. You call the shots. You're in charge of the lighting, the camera angles, the mood, everything.

To create a great prompt, you need a couple specific elements. The more you add, the better your final image will be.

First, your **Subject**. Be specific. Don't say "a dog." Say "A fluffy, golden retriever puppy, tilting its head curiously."

Next, the **Style**. What should it look like? A "hyperrealistic photograph," a "whimsical watercolor painting," "3D animation," a "vintage sci-fi poster."

Then, **Framing**. Where's the camera? Is it a "dramatic close-up shot," an "epic wide-angle landscape," a "shot from a low angle looking up"?

Then, the **Vibe**. This is all about **Lighting and Color**. "Soft, golden hour light," "moody, cinematic blue tones," "bright, saturated neon colors."
Finally, the **Details**. This is the magic. "Raindrops clinging to the window," "subtle smile," "worn leather texture on the jacket."

(SLIDE 7: Live Demo)

Tama: Alright, enough talk. Let’s actually make something.

Katie: Finally! My favorite part.

Tama: let's make a picture together. We could do it two ways. Is there something that you actually want to make a logo for? Alternatively, I'll just have you pick a random noun and verb.

Katie:

Now choose a style: **Artistic & Illustration Styles,** Experimental & Conceptual Styles, Photography & Realism Styles, Cinematic & Genre Styles

Artistic & Illustration Styles

- Studio Ghibli (whimsical, hand-painted, soft color palettes)
- Pixar 3D (bright, cinematic lighting, stylized realism)
- Digital painting (rich color, detailed brushwork)
- Watercolor illustration (soft edges, pastel tones)
- Pencil sketch / line art (minimalist, high-contrast drawing style)
- Storybook / children's illustration (playful, simplified characters)
- Surrealism (dreamlike, bending logic and scale)
- Impressionism (visible brush strokes, light and color emphasis)
- Concept art (cinematic composition, dramatic lighting)
- Low-poly 3D (geometric simplicity, stylized shapes)

📸 Photography & Realism Styles

- Hyperrealistic photography (lifelike textures and lighting)
- Cinematic realism (movie still composition, lens blur, shallow depth of field)
- Documentary / photojournalism (natural lighting, candid feel)
- Macro photography (extreme close-up details)
- Black and white film (high contrast, moody lighting)
- Vintage Polaroid (washed tones, nostalgic framing)
- Drone photography (aerial perspective, wide landscapes)
- Architectural photography (sharp geometry, clean lines)

🎬 Cinematic & Genre Styles

- Film noir (moody, high-contrast lighting, shadows)
- Cyberpunk (neon, futuristic, rainy cityscapes)
- Steampunk (Victorian tech aesthetic, brass and gears)
- Fantasy (epic, ethereal, mythic tone)
- Sci-fi realism (hard tech, futuristic architecture)
- Post-apocalyptic (dusty, muted palette, decay textures)
- Western (sun-bleached, dusty light)
- Horror / Gothic (dark atmosphere, dramatic shadows)
- Romantic comedy (bright tones, soft focus, warm light)
- Historical drama (period accuracy, muted colors)

🧠 Experimental & Conceptual Styles

- Data visualization / infographic art
- Glitch art (pixelation, distortion, abstract error textures)
- Generative / AI collage (fusion of multiple styles)
- Minimalism (flat color, clean composition, visual silence)
- Abstract geometry (shapes, lines, and gradients)
- Retrofuturism (1950s vision of the future)
- Vaporwave / synthwave (neon gradients, nostalgic digital feel)

For **Framing**, a wide shot showing the entire bookstore drifting serenely through the sky.

Framing Basics (Camera Distance & Angle)

- **Wide shot** — captures the entire scene or landscape; establishes context and setting.

- **Medium shot** — focuses on the main subject and immediate surroundings.

- **Close-up** — highlights details or emotion (e.g., a single object, face, or texture).
- **Extreme close-up** — emphasizes fine details (e.g., the texture of a page, a cat's eye).
- **Bird's-eye view / aerial** — looks straight down; great for scale and geometry.
- **Worm's-eye view / low angle** — looks up; makes the subject feel grand or imposing.
- **Eye-level shot** — natural, human perspective; feels balanced and familiar.
- **Over-the-shoulder shot** — implies a point of view or narrative; feels cinematic.
- **Tilted / Dutch angle** — dynamic, adds tension or surreal energy.
- **Panoramic** — wide horizontal framing, often for epic or tranquil scenes.

🏙️ Environmental & Contextual Framing

- **Establishing shot** — sets the scene before focusing on details (useful for "floating bookstore").
- **Interior shot** — focuses on the inside of a building or structure.
- **Exterior shot** — shows the subject in its environment (skyline, street, etc.).
- **Cutaway detail** — focuses on a single part of a larger whole (e.g., books fluttering off a balcony).
- **Framed within a frame** — uses windows, doors, or arches to create depth.
- **Reflection shot** — shows the scene through a mirror, puddle, or glass.
- **Silhouette framing** — outlines the subject against a strong light source.
- **Foreground focus** — a subject framed with blurred background elements for depth.
- **Background focus** — uses a small foreground element to lead the eye into

a distant scene.

🎬 Dynamic & Narrative Framing

- **Action framing** — captures motion mid-scene (e.g., books fluttering, clouds swirling).
- **Symmetrical composition** — centered, balanced, calming visual harmony.
- **Asymmetrical composition** — off-center, more energetic or cinematic.
- **Rule of thirds** — subject placed at a one-third line for natural visual flow.
- **Leading lines** — paths, bridges, or staircases guiding the viewer's gaze.
- **Depth framing** — multiple layers (foreground, midground, background) to add realism.
- **Point of view (POV)** — what a character or viewer is "seeing."
- **Tracking perspective** — implies motion (like following or circling the subject).
- **Vignette / spotlight framing** — isolates subject with lighting for emphasis.

🌥️ Creative / Experimental Framing

- **Dreamlike perspective** — hazy, surreal proportions or floating camera angle.
- **Split-screen / mirrored** — shows two perspectives at once.
- **Time-lapse frame** — same composition, but with movement implied (e.g., drifting clouds).
- **Macro-to-micro** — zoomed-out world fading into intricate detail.
- **Impossible perspective** — Escher-like or physics-defying angle (common in surreal AI art).

For **Vibe**, warm golden sunlight breaking through fluffy white clouds, illuminating little dust motes in the air.

And for **Details** — books spilling out of the windows, and a small, friendly cat napping on the porch.

Tama: Okay, let's send it off. This should just take a second.

("Generating..." animation plays.)

Tama: And... there we go.

(The final image of the floating bookstore fills the screen.)

Tama: the best part about this is now that we have this image, we can turn it into a video. Let's animate it.